

Table 1: Example VMs and their SELinux labels

Name	Virtual Machine process label	Virtual Machine image label
Virtual Machine 1	system_u:system_r:svirt_t:s0:c514,c932	system_u:object_r:svirt_image_t:s0:c514,c932
Virtual Machine 2	system_u:system_r:svirt_t:s0:c101,c230	system_u:object_r:svirt_image_t:s0:c101,c230

readable formats. The following snippet shows the typical contents of the `setrans.conf` file:

```
s0:c0=CompanyConfidential
s0:c1=PatentRecord
s0:c2=Unclassified
s0:c3=TopSecret
```

From the above snippet, the left hand side of the 'equals to' sign is the MCS security level, with the corresponding human readable value on the right. To list the current categories defined on your system, use the `chcat -L` command.

Each VM has unique MCS categories, and the SELinux security policy does not allow interaction between different categories, irrespective of their actual values. The file systems' device/image for each VM is also labelled with the VM's MCS label, to prevent VMs from accessing each other's resources.

Till Fedora Core 5, the typical SELinux context consisted of only three fields: `USER:ROLE:TYPE`.

Later on, SELinux added one more field, so currently it's: `USER:ROLE:TYPE:MCS Label`.

Let's look at the example of Table 1, where libvirt creates two virtual machines with dynamically generated SELinux labels.

You can obtain the SELinux labels for a running VM process, as follows:

```
# ps -eZ | grep qemu
system_u:system_r:svirt_t:s0:c514,c932
```

You can obtain the SELinux labels for the image, as follows:

```
# ls -lZ /var/lib/libvirt/qemu/images/f12.img
```

```
system_u:object_r:svirt_image_t:s0:c514,c932
```

In the above snippets, take a look at the MCS labels. Here, `s0` indicates the 'Sensitivity Level', and `c514` as well as `c932` indicate the 'category of the data'. Administrators can define these categories.

If you attempt to change the SELinux context (using the `chcon` command) of the VM image file, it will throw up SELinux AVC (Access Vector Cache) denials.

SELinux prevents virtual machine 1 (`system_u:system_r:svirt_t:s0:c514,c932`) from accessing virtual machine 2's image file (`system_u:object_r:svirt_image_t:s0:c101,c230`)—the VMs can't access each other's data, either on-disk or in-memory and hence cannot attack each other. 

References and acknowledgements

- Daniel Walsh: [danwalsh.livejournal.com \(svirt blog posts/presentations/inputs\)](http://danwalsh.livejournal.com/(svirt%20blog%20posts/presentations/inputs))
- James Morris: [http://blog.namei.org \(svirt blog posts/presentations/inputs\)](http://blog.namei.org/(svirt%20blog%20posts/presentations/inputs))
- Amit Shah and rest of Red Hat virtualization team: [Review](http://www.redhat.com/en/tech/origins/whitepapers/123456)
- Svirt Wiki: <http://selinuxproject.org/page/Svirt>
- Libvirt Project page: <http://wiki.libvirt.org>

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